

Pruning of Decomposition-Inducing GBA Transitions

接受进度一致条件、定义重构与证明替换方案

基于当前稿件 bare_jrnl_new_sample4(1).pdf 第 6-7 页的
Pruning of Decomposition-Inducing Transitions 部分

目标：将 “accepting-set union” 判断改为精确的 acceptance-progress consistency,
并将 decomposition-inducing transition 定义为可在 GBA 层直接检验的两个条件。

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1 修改结论

当前写法存在两个核心问题：第一，Definition 7 先用“转换后 NBA transition 可分解”定义 GBA transition，随后 Lemma 1 又证明“该 GBA transition 会导出可分解 NBA transition”，形成近似循环定义；第二，

$$\text{Fin}(e_{ij}) = \text{Fin}(e_{ik}) \cup \text{Fin}(e_{kj})$$

只比较接受集合的并集，没有比较 GBA-to-NBA 退广义化过程中从每一个当前进度索引出发所得到的终止进度。接受集合相同并不意味着接受进度更新相同，因为进度更新与接受集合的规定顺序以及两条边的先后次序有关。

推荐将 Definition 7 改成一个完全在 GBA 层可计算的定义：直接 GBA transition 必须同时满足

(i) **label-decomposition condition**;

(ii) **acceptance-progress consistency condition**。

随后用一个单独的 Lemma 证明：满足这两个条件的 GBA transition 所诱导的每一条 NBA transition 都是 decomposable。

2 为什么原来的接受集合并集条件不够

2.1 退广义化关心的是有序接受进度，而不是无序集合

设 GBA 的接受 transition 集合为

$$F_g = \{F_{g,1}, \dots, F_{g,m_g}\}.$$

退广义化得到的 NBA state 不仅记录 GBA state，还记录当前接受进度：

$$q_B = (q_g, \ell), \quad \ell \in \mathcal{I}_F := \{0, 1, \dots, m_g\}.$$

其中， ℓ 表示按照规定顺序已经推进到的接受集合索引； $\ell = m_g$ 对应接受 copy。因而，一条 GBA transition 的作用不是简单地“命中了哪些接受集合”，而是一个从当前进度到下一进度的映射。

原条件只要求直接边与两跳路径命中的接受集合并集相同。然而，集合并集会丢失两条 witness transition 的先后次序，也不会体现当前进度 ℓ 。因此它不能保证两条路径到达相同的 NBA copy。

2.2 一个最小反例

令 $m_g = 2$ ，当前进度为 $\ell = 0$ ，并设

$$\text{Acc}(e_{ij}^g) = \{1, 2\}, \quad \text{Acc}(e_{ik}^g) = \{2\}, \quad \text{Acc}(e_{kj}^g) = \{1\}.$$

显然，原条件成立：

$$\text{Acc}(e_{ij}^g) = \text{Acc}(e_{ik}^g) \cup \text{Acc}(e_{kj}^g) = \{1, 2\}.$$

但是, 直接边一次命中按顺序需要的 $F_{g,1}$ 和 $F_{g,2}$, 可把进度从 0 推进到 2; 两跳路径的第一条边只命中 $F_{g,2}$, 由于当前首先等待的是 $F_{g,1}$, 进度仍为 0, 第二条边再命中 $F_{g,1}$, 最终只能推进到 1。于是

$$\eta(0, e_{ij}^g) = 2, \quad \eta(\eta(0, e_{ik}^g), e_{kj}^g) = 1.$$

两者到达不同的 NBA state。该例说明: **接受集合并集相等不是接受进度一致的充分条件。**

3 建议采用的符号与精确定义

3.1 符号统一

表 1: 本小节建议采用的符号

符号	含义
e_{ij}^g	GBA transition, 避免与 NBA transition 混用同一个 e_{ij}
e_{ij}^B	NBA transition
$F_g = \{F_{g,1}, \dots, F_{g,m_g}\}$	GBA 的 transition-based accepting sets
$\text{Acc}(e^g)$	transition e^g 所属接受集合的索引集
$\mathcal{I}_F = \{0, \dots, m_g\}$	退广义化的接受进度索引集合
$\eta(\ell, e^g)$	从进度 ℓ 穿过 e^g 后的精确接受进度

建议不要再使用 “final-set consistency”。这里比较的对象不是 final states, 也不是接受集本身, 而是退广义化后的 **acceptance progress**。因此统一使用:

- acceptance-set membership: 描述 e^g 属于哪些 $F_{g,h}$;
- acceptance-progress update: 描述 $\ell \mapsto \eta(\ell, e^g)$;
- acceptance-progress consistency: 比较直接边与两跳路径的更新结果。

3.2 精确的接受进度更新函数

对任意 GBA transition $e^g \in \rightarrow_g$, 定义其接受集合索引为

$$\text{Acc}(e^g) := \{h \in \{1, \dots, m_g\} \mid e^g \in F_{g,h}\}.$$

为覆盖 LTL2BA 在接受 copy 离开时将进度重新从 0 开始检查的情况, 先定义

$$\bar{\ell} = \begin{cases} 0, & \ell = m_g, \\ \ell, & 0 \leq \ell < m_g. \end{cases}$$

然后定义

$$\eta(\ell, e^g) := \max \left\{ r \in \{\bar{\ell}, \dots, m_g\} \mid \{\bar{\ell} + 1, \dots, r\} \subseteq \text{Acc}(e^g) \right\}.$$

该定义与 LTL2BA 的 GBA-to-NBA 退广义化逻辑一致：从当前等待的下一个 **accepting set** 开始，只要 transition 连续属于后续 **accepting sets**，就依次推进；若当前已经位于接受 copy，则先重置为 0，再处理该 transition。

当前稿件中的

$$\text{Next}(\ell, \text{Fin}(e)) = \max\{r \in \{\ell, \dots, m_g\} \mid \{\ell + 1, \dots, r\} \subseteq \text{Fin}(e)\}$$

没有明确处理 $\ell = m_g$ 时的回绕。建议正文直接使用上面的 η ；这样证明与实现都不再依赖一个可能缺失回绕分支的简化公式。

4 实际算法中如何快速判断进度一致

4.1 预计算 progress-update table

不需要先构造完整 NBA。对每条 GBA transition e^g ，预计算长度为 $m_g + 1$ 的数组

$$U[e^g][\ell] := \eta(\ell, e^g), \quad \ell = 0, 1, \dots, m_g.$$

这样，直接边与两跳 witness 的进度一致性只需检查

$$U[e_{kj}^g][U[e_{ik}^g][\ell]] = U[e_{ij}^g][\ell], \quad \forall \ell \in \mathcal{I}_F.$$

也可以把上述数组称为 transition 的 *progress signature*:

$$p(e^g) = (U[e^g][0], U[e^g][1], \dots, U[e^g][m_g]).$$

两跳路径的 signature 通过函数复合得到，而不是通过接受集合并集得到。

4.2 推荐伪代码

Algorithm 1 Acceptance-progress consistency test

Require: Direct transition e_{ij}^g , witness transitions e_{ik}^g, e_{kj}^g , precomputed table U

Ensure: true iff the direct edge and the two-hop path have identical progress updates

```

1: for each  $\ell \in \mathcal{I}_F$  do
2:    $\ell_{\text{two}} \leftarrow U[e_{kj}^g][U[e_{ik}^g][\ell]]$ 
3:    $\ell_{\text{dir}} \leftarrow U[e_{ij}^g][\ell]$ 
4:   if  $\ell_{\text{two}} \neq \ell_{\text{dir}}$  then
5:     return false
6:   end if
7: end for
8: return true

```

一次性预计算的复杂度为

$$O(|\rightarrow_g| m_g),$$

每个候选 witness 的进度一致性检查为

$$O(m_g).$$

因此, 如果原 ST/decomposition 搜索为 $O(n_G d_G^3)$, 严格写法应更新为

$$O(n_G d_G^3 m_g)$$

外加一次 $O(|\rightarrow_g| m_g)$ 的预计算。若论文把 m_g 视为由固定公式决定的小常数, 可以在文字中说明这一点, 但不建议在没有说明的情况下直接省略该因子。

5 具体怎么修改当前稿件

5.1 删除或替换的内容

对当前稿件第 6-7 页, 建议按以下顺序修改:

1. 保留“为什么要移除同步复合 transition”的动机段, 但压缩为一段。
2. 保留 NBA decomposable transition 的定义, 同时把 NBA state 写成 $q_{B,i}$, 把 transition 写成 e_{ij}^B , 避免与 GBA transition 混淆。
3. 删除当前 Definition 7 中“如果转换后的 NBA transition 是 decomposable, 则称 GBA transition 为 decomposition-inducing”的循环式定义。
4. 删除整段“final set consistency condition”及

$$\text{Fin}(e_{ij}) = \text{Fin}(e_{ik}) \cup \text{Fin}(e_{kj}).$$

5. 在新 Definition 7 之前引入 $\text{Acc}(e^g)$ 、 $\mathcal{I}_F$ 和 $\eta(\ell, e^g)$ 。
6. 新 Definition 7 直接写两个可计算条件: label decomposition 与 acceptance-progress consistency。
7. 用新的 Lemma 1 证明这两个 GBA-level 条件足以导出 NBA decomposability。
8. 在删除 direct transition 前补一句: witness transitions 必须保留在 pruned GBA 中。

5.2 逻辑结构应改成下面的顺序

动机 $\rightarrow$ NBA decomposable transition 定义 $\rightarrow$ GBA 接受进度更新 $\rightarrow$ GBA decomposition-inducing transition 的两个条件 $\rightarrow$ 快速判定方法 $\rightarrow$ Lemma: 两个条件推出 NBA decomposable。

这比当前“先用 NBA 可分解性定义 GBA transition, 再证明它在 NBA 中可分解”的结构更严密, 也更突出本文实际完成的工作: 在完整 NBA 尚未生成时, 仅利用 GBA transition、accepting-set membership 和退广义化进度更新, 就能识别将来会形成可分解 NBA transition 的 direct GBA transition。

6 修改后的英文正文（可直接替换）

Rewritten subsection

2) Pruning of Decomposition-Inducing GBA Transitions: The second pruning step is performed after the GBA has been constructed. It targets direct transitions whose labels combine the requirements of a two-hop path and therefore encode an avoidable simultaneous execution of otherwise sequentially realizable subtasks. The main difficulty is that the pruning is performed before the complete NBA is available. Hence, a direct GBA transition can be removed only when its two-hop witness preserves both the transition-label decomposition and the acceptance progress used in the subsequent GBA-to-NBA degeneralization.

Definition 6 (Decomposable NBA transition). Consider an NBA transition

$$e_{ij}^B = (q_{B,i}, \sigma_{ij}, q_{B,j}) \in \rightarrow_B.$$

The transition e_{ij}^B is called decomposable if there exist an NBA state $q_{B,k}$ and two input symbols $\sigma_{ik}, \sigma_{kj} \in \Sigma_B$ such that

$$(q_{B,i}, \sigma_{ik}, q_{B,k}) \in \rightarrow_B, \quad (q_{B,k}, \sigma_{kj}, q_{B,j}) \in \rightarrow_B,$$

and

$$\sigma_{ij} = \sigma_{ik} \cup \sigma_{kj}.$$

To identify such transitions directly in the GBA, let

$$F_g = \{F_{g,1}, \dots, F_{g,m_g}\}$$

be the collection of accepting transition sets and define the acceptance-set index set of a GBA transition $e^g \in \rightarrow_g$ as

$$\text{Acc}(e^g) := \{h \in \{1, \dots, m_g\} \mid e^g \in F_{g,h}\}.$$

Each state of the degeneralized NBA is represented by (q_g, ℓ) , where $q_g \in Q_g$ and $\ell \in \mathcal{I}_F := \{0, \dots, m_g\}$ is the current acceptance-progress index. Let $\eta(\ell, e^g)$ denote the progress index obtained after traversing e^g from progress ℓ . Specifically, let $\bar{\ell} = 0$ if $\ell = m_g$ and $\bar{\ell} = \ell$ otherwise, and define

$$\eta(\ell, e^g) = \max \left\{ r \in \{\bar{\ell}, \dots, m_g\} \mid \{\bar{\ell} + 1, \dots, r\} \subseteq \text{Acc}(e^g) \right\}.$$

Definition 7 (Decomposition-inducing GBA transition). Consider a direct GBA transition

$$e_{ij}^g = (q_{g,i}, \sigma_{ij}, q_{g,j}) \in \rightarrow_g.$$

The transition e_{ij}^g is called decomposition-inducing if there exist a GBA state $q_{g,k} \in Q_g$ and two GBA transitions

$$e_{ik}^g = (q_{g,i}, \sigma_{ik}, q_{g,k}) \in \rightarrow_g, \quad e_{kj}^g = (q_{g,k}, \sigma_{kj}, q_{g,j}) \in \rightarrow_g,$$

such that the following two conditions hold:

1) *Label-decomposition condition*:

$$\sigma_{ij} = \sigma_{ik} \cup \sigma_{kj}.$$

2) *Acceptance-progress consistency*:

$$\eta(\eta(\ell, e_{ik}^g), e_{kj}^g) = \eta(\ell, e_{ij}^g), \quad \forall \ell \in \mathcal{I}_F.$$

The pair (e_{ik}^g, e_{kj}^g) is called a two-hop decomposition witness of e_{ij}^g .

Efficient implementation. The acceptance-progress condition can be checked without constructing the complete NBA. For every GBA transition e^g , the implementation precomputes the table entry $U[e^g][\ell] = \eta(\ell, e^g)$ for all $\ell \in \mathcal{I}_F$. A candidate witness then satisfies the progress condition if and only if

$$U[e_{kj}^g][U[e_{ik}^g][\ell]] = U[e_{ij}^g][\ell]$$

for every $\ell \in \mathcal{I}_F$. Thus, one witness can be verified using $m_g + 1$ table lookups and comparisons.

Lemma 1. Let e_{ij}^g be a decomposition-inducing GBA transition with a two-hop witness (e_{ik}^g, e_{kj}^g) . Then every NBA transition induced by e_{ij}^g is decomposable according to Definition 6.

Proof. Fix an arbitrary acceptance-progress index $\ell \in \mathcal{I}_F$. In the GBA-to-NBA degeneralization, the direct GBA transition e_{ij}^g induces the NBA transition

$$((q_{g,i}, \ell), \sigma_{ij}, (q_{g,j}, \eta(\ell, e_{ij}^g))).$$

The two witness transitions induce the two-hop NBA path

$$(q_{g,i}, \ell) \xrightarrow{\sigma_{ik}} (q_{g,k}, \eta(\ell, e_{ik}^g)) \xrightarrow{\sigma_{kj}} (q_{g,j}, \eta(\eta(\ell, e_{ik}^g), e_{kj}^g)).$$

By the acceptance-progress consistency condition,

$$\eta(\eta(\ell, e_{ik}^g), e_{kj}^g) = \eta(\ell, e_{ij}^g),$$

so the direct transition and the two-hop path reach the same terminal NBA state. Moreover, the label-decomposition condition gives $\sigma_{ij} = \sigma_{ik} \cup \sigma_{kj}$. Therefore, the NBA transition induced by e_{ij}^g is decomposable according to Definition 6. Since ℓ was arbitrary, the conclusion holds for every NBA transition induced by e_{ij}^g . $\square$ Accordingly, a decomposition-inducing direct transition can be marked for pruning before NBA construction, provided that its two-hop witness transitions are retained in the pruned GBA.

6.1 关于 transition label 的一个必要说明

上面的可直接替换稿沿用了当前论文的

$$\sigma_{ij} = \sigma_{ik} \cup \sigma_{kj}$$

以保持最小改动。若代码中的 transition label 实际上是包含正、负文字的符号 guard，而不是“需要同时为真的正命题集合”，则正文应改写为

$$\gamma_{ij} \equiv \gamma_{ik} \wedge \gamma_{kj},$$

或分别比较 positive/negative literal bitsets。否则，集合并集无法完整表达带否定命题的逻辑合取。该问题独立于接受进度修改，但审稿人可能据此质疑 label decomposition 的语义。

7 可直接粘贴的 LaTeX

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\textbf{2) Pruning of Decomposition-Inducing GBA Transitions:}
The second pruning step is performed after the GBA has been
constructed. It targets direct transitions whose labels combine
the requirements of a two-hop path and therefore encode an
avoidable simultaneous execution of otherwise sequentially
realizable subtasks. The main difficulty is that the pruning is
performed before the complete NBA is available. Hence, a direct
GBA transition can be removed only when its two-hop witness
preserves both the transition-label decomposition and the
acceptance progress used in the subsequent GBA-to-NBA
degeneralization.
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\begin{definition}[Decomposable NBA transition]
Consider an NBA transition
\begin{equation}
e^B_{ij} = (q_{B,i}, \sigma_{ij}, q_{B,j}) \in \text{rightarrow}_B.
\end{equation}
The transition  $e^B_{ij}$  is called decomposable if there exist
an NBA state  $q_{B,k}$  and two input symbols
 $\sigma_{ik}, \sigma_{kj} \in \Sigma_B$  such that
\begin{equation}
(q_{B,i}, \sigma_{ik}, q_{B,k}) \in \text{rightarrow}_B,
\quad
(q_{B,k}, \sigma_{kj}, q_{B,j}) \in \text{rightarrow}_B,
\end{equation}
and
\begin{equation}
\sigma_{ij} = \sigma_{ik} \cup \sigma_{kj}.
\end{equation}
\end{definition}
```

To identify such transitions directly in the GBA, let
 $F_g = \{F_{g,1}, \dots, F_{g,m_g}\}$ be the collection of

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accepting transition sets and define
\begin{equation}
\operatornamename{Acc}(e^g)
:= \{h \in \{1, \ldots, m_g\} \mid e^g \in F_{\{g, h\}}\}.
\end{equation}
Each state of the degeneralized NBA is represented by
 $(q_g, \ell)$ , where
 $\ell \in \mathcal{I}_F := \{0, \ldots, m_g\}$  is the current
acceptance-progress index. Let  $\eta(\ell, e^g)$  denote the
progress index obtained after traversing  $e^g$  from progress
 $\ell$ . Specifically, let  $\bar{\ell} = 0$  if  $\ell = m_g$  and
 $\bar{\ell} = \ell$  otherwise, and define
\begin{equation}
\eta(\ell, e^g)
=
\max \Bigl\{
r \in \{\bar{\ell}, \ldots, m_g\}
\mid \eta(\bar{\ell} + 1, r) \in \operatornamename{Acc}(e^g)
\Bigr\}.
\end{equation}

\begin{definition}[Decomposition-inducing GBA transition]
Consider a direct GBA transition
\begin{equation}
e^g_{\{ij\}} = (q_{\{g, i\}}, \sigma_{\{ij\}}, q_{\{g, j\}})
\in \rightarrow_g.
\end{equation}
The transition  $e^g_{\{ij\}}$  is called decomposition-inducing if
there exist a GBA state  $q_{\{g, k\}} \in Q_g$  and two GBA
transitions
\begin{equation}
e^g_{\{ik\}} = (q_{\{g, i\}}, \sigma_{\{ik\}}, q_{\{g, k\}}) \in \rightarrow_g,
\quad
e^g_{\{kj\}} = (q_{\{g, k\}}, \sigma_{\{kj\}}, q_{\{g, j\}}) \in \rightarrow_g,
\end{equation}
such that the following two conditions hold:
\begin{enumerate}
\item \emph{Label-decomposition condition:}
\begin{equation}
\sigma_{\{ij\}} = \sigma_{\{ik\}} \cup \sigma_{\{kj\}}.
\end{equation}
\item \emph{Acceptance-progress consistency:}
\begin{equation}
\eta(\ell, e^g_{\{ik\}}) \leq \eta(\ell, e^g_{\{kj\}}) \leq \eta(\ell, e^g_{\{ij\}}),
\quad \forall \ell \in \mathcal{I}_F.
\end{equation}
\end{enumerate}

```

The pair (e^g_{ik}, e^g_{kj}) is called a two-hop decomposition witness of e^g_{ij} .

`\end{definition}`

`\begin{remark}[Efficient implementation]`

For every GBA transition e^g , precompute

$U[e^g][\ell] = \eta(\ell, e^g)$ for all

$\ell \in \mathcal{I}_F$. A candidate witness satisfies the progress condition if and only if

`\begin{equation}`

$U[e^g_{kj}] \bigl[U[e^g_{ik}][\ell] \bigr] \bigr]$

$= U[e^g_{ij}][\ell],$

`\quad \forall \ell \in \mathcal{I}_F.`

`\end{equation}`

Thus, one witness can be verified using m_g+1 table lookups and comparisons.

`\end{remark}`

`\begin{lemma}`

Let e^g_{ij} be a decomposition-inducing GBA transition with a two-hop witness (e^g_{ik}, e^g_{kj}) . Then every NBA transition induced by e^g_{ij} is decomposable.

`\end{lemma}`

`\begin{proof}`

Fix an arbitrary $\ell \in \mathcal{I}_F$. The direct GBA transition e^g_{ij} induces the NBA transition

`\begin{equation}`

$\bigl((q_{g,i}, \ell), \sigma_{ij}, (q_{g,j}, \eta(\ell, e^g_{ij})) \bigr).$

`\end{equation}`

The witness transitions induce the two-hop NBA path

`\begin{equation}`

$(q_{g,i}, \ell)$

$\xrightarrow{\sigma_{ik}}$

$(q_{g,k}, \eta(\ell, e^g_{ik}))$

$\xrightarrow{\sigma_{kj}}$

$\left(q_{g,j}, \right.$

$\left. \eta(\left(\eta(\ell, e^g_{ik}), e^g_{kj} \right) \right) \right).$

`\end{equation}`

By acceptance-progress consistency, the direct transition and the two-hop path reach the same terminal NBA state. Moreover, $\sigma_{ij} = \sigma_{ik} \cup \sigma_{kj}$ by the label-decomposition condition. Hence, the NBA transition induced by e^g_{ij} is decomposable. Since ℓ was arbitrary, the conclusion holds for every NBA transition induced by e^g_{ij} .

`\end{proof}`

Accordingly, a decomposition-inducing direct transition can

be marked for pruning before NBA construction, provided that its two-hop witness transitions are retained in the pruned GBA.

8 需要同步修改的其他位置

8.1 Lemma 2 的引用句

将当前

By Lemma 1, every decomposition-inducing transition removed in the GBA corresponds to a decomposable transition in the NBA.

改为更精确的表述：

By Lemma 1, every NBA transition induced by a pruned decomposition-inducing GBA transition admits a retained two-hop replacement that reaches the same terminal NBA state and preserves the transition-label decomposition.

这样能够明确 Lemma 1 证明的是“每个进度 copy 上的 induced NBA transition 都存在相同终点的两跳替代”，而不是含糊地说一条 GBA transition “对应”一条 NBA transition。

8.2 复杂度分析

将

The proposed ST pruning checks whether a direct transition can be replaced by a two-hop transition while preserving the corresponding acceptance progress. Its additional worst-case cost is $O(n_G d_G^3)$.

建议改为：

For each candidate direct transition and two-hop witness, the proposed pruning compares their precomputed acceptance-progress signatures over all $m_g + 1$ degeneralization indices. After an $O(|\rightarrow_g| m_g)$ preprocessing step, one witness check requires $O(m_g)$ time. Hence, the additional worst-case cost is $O(n_G d_G^3 m_g)$, or $O(n_G d_G^3)$ when m_g is treated as a fixed formula-dependent constant.

8.3 全文术语统一

全文检索并统一如下：

- “final-set consistency” $\rightarrow$ “acceptance-progress consistency”;
- “final set of a transition” $\rightarrow$ “acceptance-set index set of a transition”;
- GBA transition 用 e^g , NBA transition 用 e^B ;
- 不再使用未在 Definition 4 中定义的 δ_g , 统一通过 $e^g \in \rightarrow_g$ 表示 GBA transition;
- “accepting state consistency” 不准确, 应写 “acceptance-progress consistency” 或 “the same terminal NBA copy”。

9 审稿视角下仍需保留的一条边界说明

新的 Definition 7 和 Lemma 1 能严格证明: 两个 GBA-level 条件足以保证直接边在每个退广义化进度 copy 上都诱导一条 decomposable NBA transition。它**不能单独证明**任意删除顺序都保持语言或最优代价。实际 pruning 时, 必须保证 witness transitions e_{ik}^g 与 e_{kj}^g 在最终 pruned GBA 中仍然保留; 否则后续递归删除可能把替代路径一并删掉。

最小改动做法是在删除 e_{ij}^g 时记录 witness pair, 并仅当两条 witness transition 已确定保留时执行删除。若后文继续声称 cost non-increasing 或全局 optimality preserving, 还需要独立给出代价层面的证明; 本次“接受进度一致”修改只解决 automaton-state equivalence。

10 最终推荐

建议采用本文给出的完整替换稿。其核心判断条件为

$$\eta\left(\eta(\ell, e_{ik}^g), e_{kj}^g\right) = \eta(\ell, e_{ij}^g), \quad \forall \ell \in \mathcal{I}_F$$

而不是接受集合并集相等。新 Definition 7 直接以 “label decomposition + acceptance-progress consistency” 定义 decomposition-inducing GBA transition; 新 Lemma 1 再从任意 ℓ 出发, 证明两跳路径与直接边到达同一个 terminal NBA state, 因此导出 NBA decomposability。该结构既消除了循环定义, 也把本文实际的 GBA-level 判定机制表达得更清楚。

参考依据

- [1] 用户当前稿件 *bare_jrnl_new_sample4(1).pdf*, 第 6-7 页, Section V-B.
- [2] P. Gastin and D. Oddoux, “Fast LTL to Büchi Automata Translation,” CAV, 2001.
- [3] LTL2BA official implementation, *buchi.c*, especially `next_final` and `make_btrans`; an accepting copy is reset to progress 0 before consecutive accepting sets are advanced.